A rapid sand filter is an open water tight chamber, 3 to 3.5 m deep. It has coarse sand filter media, 0.6 to 0.75 m thick, laid on 0.45 m thick graded gravel. The under-drainage system is supported by concrete floor. The under-drainage system consists manifold with strainers mounted on top and laterals. Laterals have perforations on sides. The filtration is effected by gravity.

The rate of filtration is about 3000 to 6000 litres/m^2/hour. Its bacterial efficiency is 80 to 90%. It removes turbidity upto 30 to 40 ppm.

A typical longitudinal section of a rapid sand filter is shown in fig. The following are its essential parts

i. Enclosure tank
ii. Filter media
iii. Base material
iv. Under-drainage system
v. Appurtenances

Action during filtration

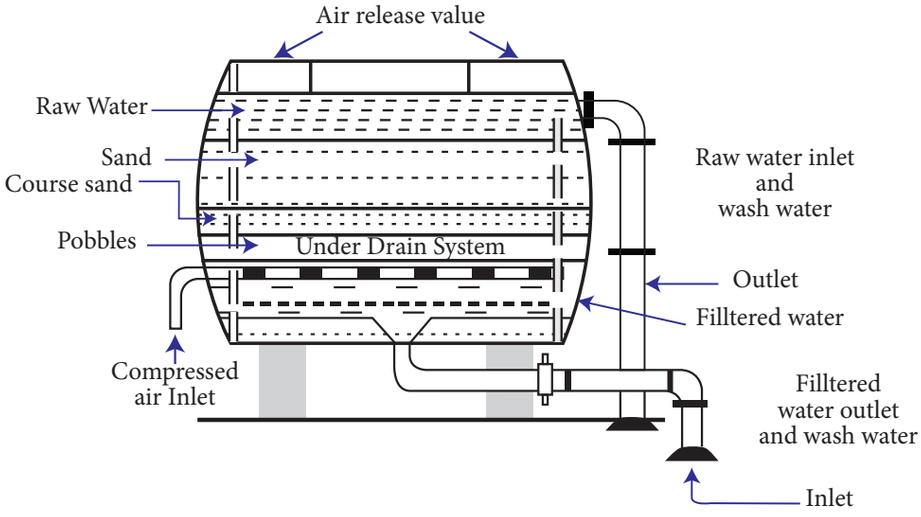
Inlet valve is opened and water from coagulated sedimentation tank is allowed to enter the filter. Effluent valve is opened to carry filtered water to clear water reservoir. During this time all other valves remains closed. Only inlet and effluent valves are opened.

Action during backwashing

Backwashing is done when the loss of head reaches the maximum permissible limit of 2.5 to 3.5 m. The filter is drained out leaving a very small depth of water standing above the filter bed. Now, the compressed air is sent under pressure through the under drainage system for about 2 to 3 minutes. This agitates the mass of water. The agitated water loosens the dirt from the surface of sand grains. Then, an upward flow of water from a high level tank is sent through the bed. This causes the sand bed to expand, agitate the sand grains and wash off the surface deposits. The deposits are carried by wash water troughs and disposes through wash water drains. During backwashing the valve positions are the inlet and effluent are closed, and the valves of wash water trough and wash water drains remains open.

3. Pressure Filters

They consist of closed watertight cylindrical metal tanks of 1.5 to 3 m diameter and length or height 3.5 to 8 m. It contains filter media as of rapid sand filter. The raw water mixed with a dose of coagulant is



Horizontal pressure filter (sectional view)

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